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Software Engineering Internship Team
SpaceX

Dear Hiring Team,

I am applying for the Summer 2027 Software Engineering Internship. I am a Computer Science and Finance student at UNC Charlotte, and the work I have done outside the classroom has mostly involved getting software to behave correctly when the physical world refuses to cooperate.

On the F1TENTH autonomous racing team I built a tenth-scale race car with a multidisciplinary group, writing ROS and Python for path planning, obstacle avoidance, and real-time sensor processing. The interesting part was not the initial implementation. It was the weeks of tuning PID control and localization against tracks that kept breaking our assumptions, and integrating LIDAR, IMU, and camera data into a stack that had to make decisions faster than we could reason about them. Most of what I learned came from the runs that failed.

I am currently a Product Engineer at VibeGrow, where I work on multi-agent coordination: the problem of getting several autonomous processes to pursue one objective without duplicating work or acting on state that has already changed. It is a different domain from vehicle control, but the underlying difficulty is familiar. Systems that operate independently still have to agree on what is true.

I also built a logistics routing tool using Dijkstra's, A*, and genetic algorithms under constraints like traffic and delivery windows, wrapped in a Flask interface so people without a technical background could test solutions themselves. That instinct, making a system legible to the people who depend on it, came from running needs discovery with clients at Tabner, and it has shaped how I document and explain the things I build.

I would like to work on problems where the consequences of getting it wrong are physical and immediate. I am available full time onsite for the summer and would welcome the chance to discuss where I might be useful.

Sincerely,
Charan Reddy Yakkanti